INTRO TO COMPUTER SCIENCE & CODING LOGIC TUTORIAL

Intro to local AI

unpacking and finding alternatives

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source: https://www.printables.com/model/1617844-openclaw-lobster

Have you heard of running AI *locally*?

A

Nope?

D

I think so

Today's plan

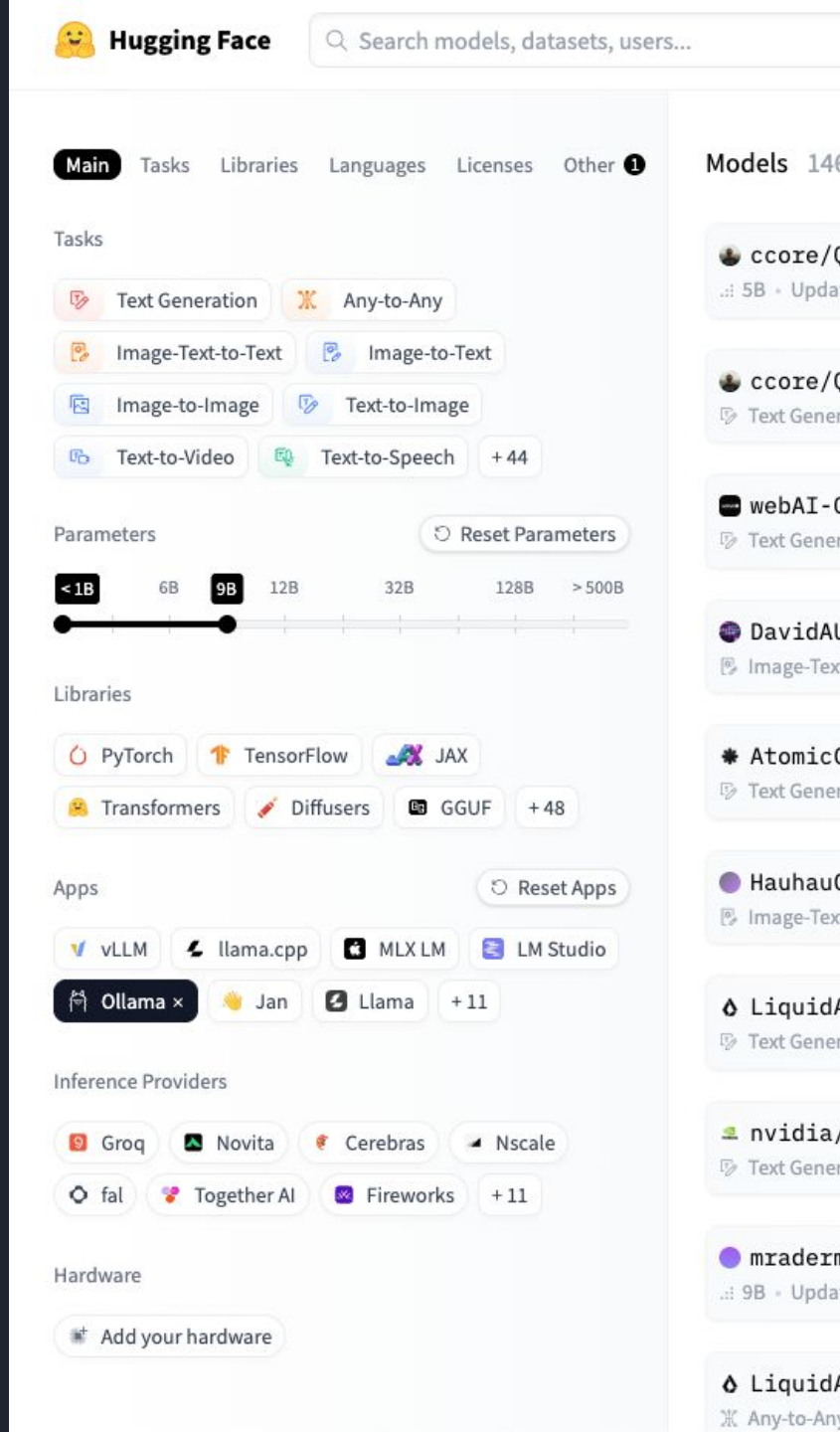
1. What *local AI* means
2. How it works
3. Live demo with Ollama + OpenClaw
4. What could you make?
5. Q&A

An AI model is a *file* you can download.

Behind every chatbot is a big file of learned numbers and a program that runs it. That file used to stay on a company's servers. Now many are free to download and run on your own computer.

`llama` **Meta** | `gemma` **Google** | `qwen` **Alibaba**

Check out https://huggingface.co



Same model, different place to do the math

Hosted

ChatGPT, Claude, Gemini, …

- **Your prompt travels** over the internet
- Someone else's **data center** runs the model
- **You pay** per token or per seat
- Privacy is a policy (**a promise**)

Local

Today's demo

- The model file **sits on your disk**
- **Your own machine** does the math
- No account, no bill, **works offline**
- Privacy is physical: **nothing leaves**

Why artists and makers choose to run AI locally

- **Your material stays yours**

  Private archives, interviews, audience data: nothing gets uploaded.

- **Works without internet**

  A gallery, a festival, a dome: the piece keeps running when the Wi-Fi doesn’t.

- **No bill per use**

  An installation can talk to visitors all day without a subscription meter running.

A local model is smaller than ChatGPT or Claude: less clever, but often good enough

Every local setup has three pieces

The model

Billions of calculations. Downloaded once.
often **.gguf** **.safetensors** **.onnx**

The program

Loads the weights into memory and does the math.
llama.cpp, Ollama, LM Studio,

How you talk to it

Terminal chat, a web UI, or an y API your code can call

What fits on your computer?

```
$ ollama run llama3.2:3b
```

Who made it **The version** **The size (3b: 3 billion learned numbers. Bigger = smarter, slower, heavier**

Memory rule of thumb: the download size ≈ the memory it needs. Leave room for your other apps.

```
gemma3:1b takes up  ~0.8 GB
llama3.2:3b  takes up · fine on an 8 GB laptop
```

canirun.ai

Not sure? It detects your computer and ranks the open models that fit.

Let’s run one

Three commands to a running model

zsh

```
# 1. install (or get the app at ollama.com/download)
$ curl -fsSL https://ollama.com/install.sh | sh

# 2. download a small model (~2 GB, once)
$ ollama pull qwen3.5:9b
# or check if you have any installed with ollama list

# 3. chat with it · type /bye to leave
$ ollama run qwen3.5:9b
# or check ollama ps (process status)
```

Our prompt: *Write a six-word poem about a room full of light.*

response: Quiet room bathed in golden light.

+ turn off wifi and ask another !

Bigger is smarter, but slower and hungrier

zsh

```
# small: ~2 GB
$ ollama run --verbose llama3.2:3b "Describe blue to someone who has never seen it."

# same question, a model about 9× bigger (~17 GB)
$ ollama run --verbose gemma3:27b "Describe blue to someone who has never seen it."
```

How fast the words appear, and whether the answer is better.

No size is “best”, only the right size for a job on a machine.

What could you make with it?

- **A talking installation**

A character that answers visitors, offline, all day.

- **An archive helper**

Tag, caption or summarize a private collection without uploading it.

- **Game dialogue**

Characters in a game or interactive story who improvise their lines.

- **A brainstorm partner**

Titles, variations, artist-statement drafts: you keep the final word.

Your tools can talk to it: Ollama runs as a small web service on your computer, so three.js and p5.js can send it text and get text back.

From a chatbot to an agent

- **Chatbot**

Text in, text out. It can only talk.

- **Agent**

It can also do things: open files, search, send messages. Tools like OpenClaw use a local model this way.

small demo (**OpenClaw**)

Because an agent can act, **only give it only the tools and access it needs**

Microsphère

A prototype presented at MUTEK FORUM, AUGUST 2026

An interactive orb that reads people's presence through depth cameras and turns it into a live, AI-generated story and visuals, projected back onto its surface.

Full stack: depth, IR and RGB cameras - Carto - LM Studio + Qwen3 - ossia score - Stream Diffusion - NDI - OSC - two RTX 4000 GPUs - projector



Cameras see visitors

Depth, infrared and colour cameras

A local model tells the story

LM Studio running Qwen3 (~6 GB), no cloud

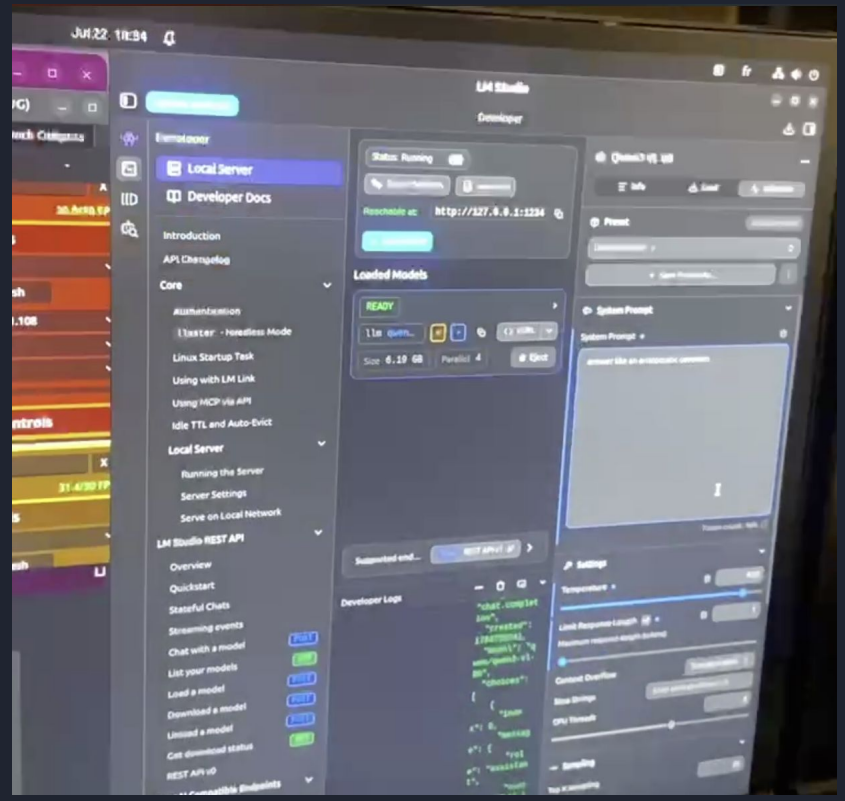
Images generated live

StreamDiffusion

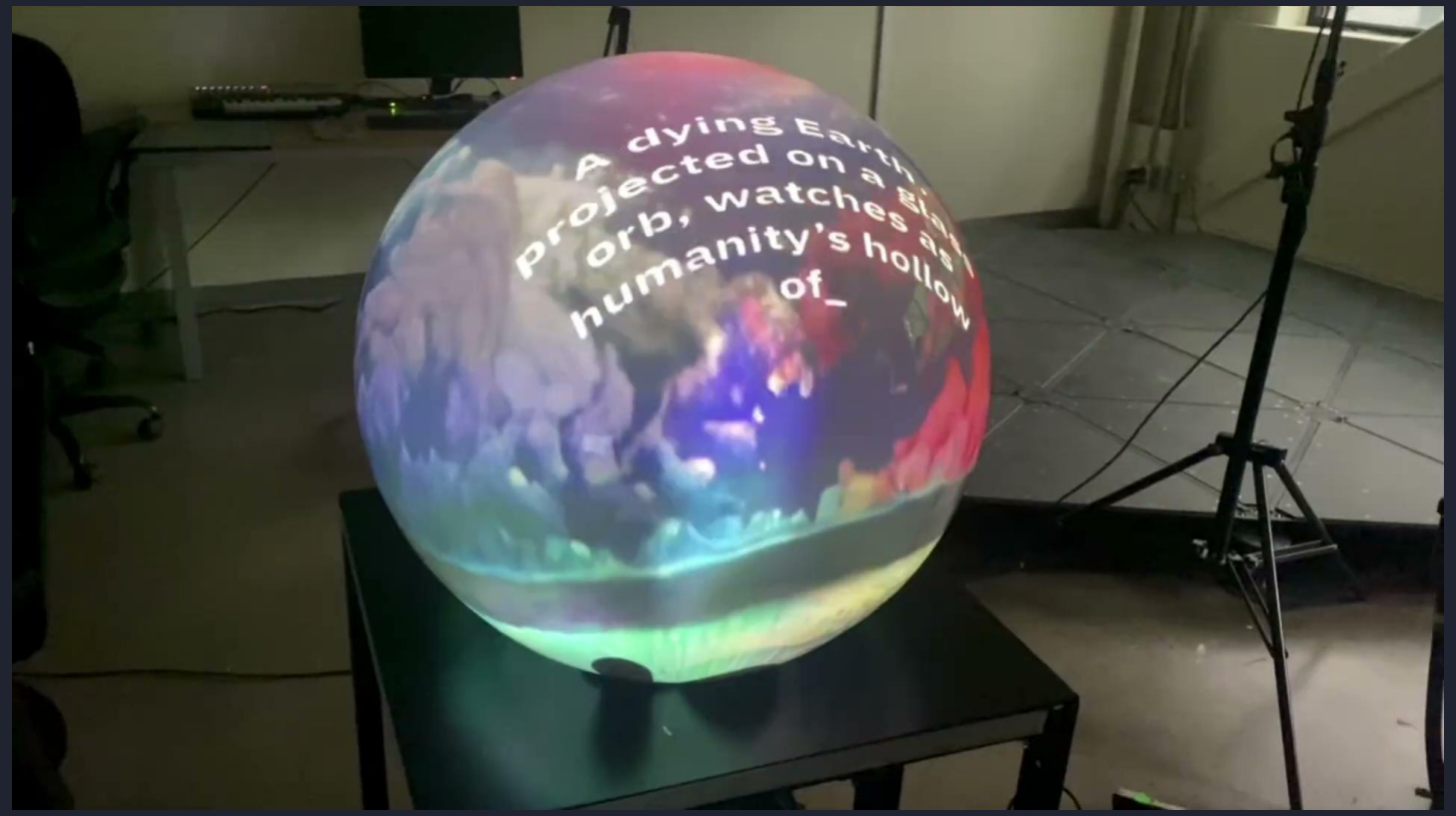
Projected onto the orb

Projector on the sphere's surface

Microsphère



string: "Hmph. A cave-dweller sees a tech-cave with a man in a hat, hunched over glowing screens like a cave-king worshipping fire. A purple orb sits like a strange, shiny berry on a box, and a tall pole with a camera watches like a sentry. The place is cluttered with wires and machines, but the man seems focused, like a cave-king tending to his sacred fire."



Is it safe?

Thank you! Any questions?



Ask anything! Now's your time: make the most of your tuition feeeeeees.

Who uses our data?

Where does our data go?